



MANIPAL SCHOOL OF INFORMATION SCIENCES

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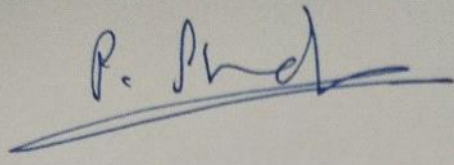
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Master of Engineering - M.E in Computer Science and Engineering (Embedded Systems)

Course Plan

Course Name	:	Data Structures and Algorithms
Course Code	:	ESE 5102
Academic Year	:	2026-2027
Semester	:	I
Name of the Course Coordinator	:	Mr. Sudhakara Upadya
Name of the Program Coordinator	:	Dr. B Dinesh Rao

	
Date:	14/07/2026
Signature of Program Coordinator	
with Date	

	
Date:	14/07/2026
Signature of Course Coordinator	
with Date	



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Program Education Objectives (PEOs)

The overall objectives of the Learning Outcomes-based Curriculum Framework (LOCF) for Master of Engineering - M.E in Computer Science and Engineering (Embedded Systems), program are as follows

PEO No.	Education Objective
PEO 1	Enable to draw upon fundamental and advanced knowledge to apply analytical and computational approaches to solve technological problems in embedded systems.
PEO 2	Introduce state of art technologies in the area of embedded systems and inculcate ethical practices to make industry-ready professionals.
PEO 3	Promote scientific and societal advancement through research and entrepreneurship.



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Program Outcomes (POs)

By the end of the postgraduate program in Master of Engineering - M.E in Computer Science and Engineering (Embedded Systems), graduates will be able to:

PO 1	An ability to independently carry out research /investigation and development work to solve practical problems.
PO 2	An ability to write and present a substantial technical report/document.
PO 3	An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program.
PO 4	An ability to develop and implement embedded systems requirements based on theoretical principles and practical knowledge.
PO 5	An ability to demonstrate knowledge of the underlying principles and evaluation methods for analyzing and decision-making.



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1. Course Plan

1.1 Primary Information

Course Name	:	Data Structures and Algorithms [ESE 5102]
L-T-P-C	:	3-0-0-3
Contact Hours	:	36 Hours
Pre-requisite	:	C Programming
Core/ PE/OE	:	Core

1.2 Course Outcomes (COs), Program outcomes (POs) and Bloom's Taxonomy Mapping

CO	At the end of this course, the student should be able to:	No.ofContact Hours	Program Outcomes (POs)	BL
CO1	Analyse the implementations of elementary data structures.	12	PO3	3
CO2	Design and implementation of data structures like trees and graphs.	15	PO4	4
CO3	Analyse the concepts of divide and conquer, dynamic programming, greedy, and back tracking techniques to efficiently solve problems.	9	PO5	4



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1.3 Assessment Plan

Components	Mid semester exam (MSE)	Flexible assessments	End semester exam (ESE) / Makeup exam
Duration	90 minutes	1 week from posting assignment	180 minutes
Weightage	0.3	0.2	0.5
Typology of questions	Apply and Analyze	Apply and Analyze	Apply and Analyze
Pattern	Each question of 3 to 10 Marks	Solve problems by applying the design principles and additional reads	10 questions of 10 marks each - each question containing subquestions or single question depending on topic
Schedule	As per academic calendar.	Assignment submission: LMS	As per academic calendar.
Topics covered	Algorithm Performance measures, List, Queue, Sorting, Search	Graphs, Sets, Algorithm Design Techniques	Comprehensive examination covering the full syllabus. Students are expected to answer all questions.



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1.4 Lesson Plan

L. No.	TOPICS	Course Outcome Addressed
L0	Course delivery plan, Course assessment plan, Course outcomes, Program outcomes, CO-PO mapping, reference books.	---
L1	Algorithm Specification	C01
L2	Performance Analysis	C01
L3	Analysis of Recursive Programs	C01
L4	Solving Recurrence Equations	C01
L5	General Solutions for a Large Class of Recurrences	C01
L6	Randomized Algorithms	C01
L7	Implementation of Lists - SLL	C01
L8	Implementation of Lists - DLL	C01
L9	Implementation of Stacks	C01
L10	Implementation of Queues	C01
L11	Implementation of Priority Queues	C01



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L12	Implementation of Circular Queues	C01
L13	Quick sort	C02
L14	Heap sort	C02
L15	Merge sort	C02
L16	Binary search	C02
L17	linear search	C02
L18	Fibonacci search	C02
L19	Trees - Basic Terminology	C02
L20	Implementation of Trees	C02
L21	Binary Trees	C02
L22	Binary Search Trees	C02
L23	Types of balanced trees, construction and its operation	C02
L24	Graphs: Basic definitions, Representation of Graphs	C02
L25	Minimum Cost Spanning Tree	C02



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L26	Single Source Shortest Paths	C02
L27	All-Pairs Shortest Path	C02
L28	Introduction to Sets, A Linked- List implementation of Set	C03
L29	The Dictionary	C03
L30	The Hash Table Data Structure	C03
L31	Divide and Conquer Algorithms	C03
L32	Dynamic Programming	C03
L33	Greedy Algorithms - Knapsack	C03
L34	Greedy Algorithms - Job Scheduling	C03
L35	Backtracking - NQueen problem	C03
L36	Backtracking - Sum of Subsets	C03



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1.5 References

1. T. H. Cormen, C. E. Leiserson, and R. L. Rivest, Introduction to Algorithms, 3rd ed., MIT Press, 2009.
2. A. V. Aho, J. E. Hopcroft, and J. D. Ullman, Data Structures and Algorithms, Addison-Wesley, 1983.
3. M. A. Weiss, Data Structures and Algorithm Analysis in C, 2nd ed., Pearson, 1997.
4. E. Horowitz, S. Sahni, and S. Rajasekaran, Computer Algorithms, 2nd ed., Silicon Press, 2007
5. MOOC: <https://in.coursera.org/learn/algorithms-on-graphs?specialization=data-structures-algorithms#enroll>

1.6 Other Resources (Online, Text, Multimedia, etc.)

1. Web Resources: Blog, Online tools and cloud resources.
2. Journal Articles.



1.7 Course Timetable

I Semester						Room: LG01-LH10			
	8-9	9-10	10-11	11-12	12-1	1-2	2-3	3-4	4-5
Monday		DS							
Tuesday									
Wednesday		DS							
Thursday									
Friday		DS							
Saturday									



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1.8 Assessment Plan

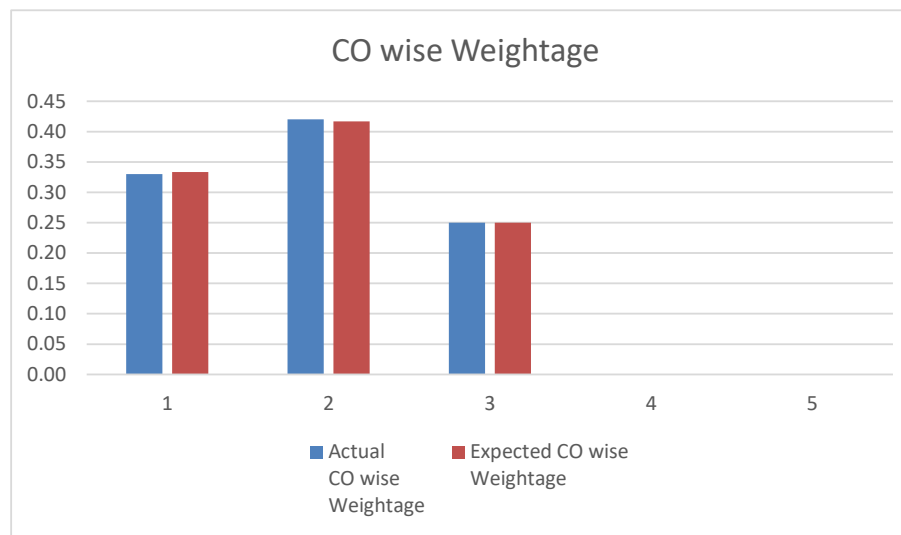
COs		Marks & Weightage				
CO No.	CO Name	MSE Marks (Max. 50)	Assignment Marks (Max. 20)	ESE Marks (Max. 100)	Actual CO wise Weightage	Expected CO wise Weightage
C01	Analyse the implementations of elementary data structures.	30		30	0.33	0.33
C02	Design and implementation of data structures like trees and graphs.	20	10	40	0.42	0.42
C03	Analyse the concepts of divide and conquer, dynamic programming, greedy, and back tracking techniques to efficiently solve problems.		10	30	0.25	0.25
	Marks (weightage)	0.3	0.2	0.5	1.00	1.00



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Note:

- In-semester Assessment is considered as the Internal Assessment (IA) in this course for 50 marks, which includes the performances in class participation, assignment work, class tests, MSE, quizzes etc.
- ESE for this course is conducted for a maximum of 100 and the same will be scaled down to 50 for grading.
- ESE marks for a maximum of 50 and IA marks for a maximum of 50 are added for a maximum of 100 marks to decide upon the grade in this course.

$$\text{Weightage for CO1} = (\text{MSE marks for CO1} / 1.6666 + \text{Assignment marks for CO1} / 1.0 + \text{ESE marks for CO1} / 2) / 100$$



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1.9 Assessment Details

The assessment tools to be used for the Current Academic Year (CAY) are as follows:

Sl. No.	Tools	Weightage	Frequency	Details of Measurement (Weightage/Rubrics/Duration, etc.)
1	MSE	0.3	1	- Performance is measured using MSE attainment level. - Reference: question paper and answer scheme. - MSE is assessed for a maximum of 50 marks and scaled down to 30 marks.
2	Assignments	0.2	2	- Performance is measured using assignments/quiz attainment level. - Assignments/quiz are evaluated for a maximum of 20 marks.
3	ESE	0.5	1	- Performance is measured using ESE attainment level. - Reference: question paper and answer scheme. - ESE is assessed for a maximum of 100 marks and scaled down to 50 marks.

1.10 Course Articulation Matrix

CO	PO1	PO2	PO3	PO4	PO5
CO1			Y		
CO2				Y	
CO3					Y
Average Articulation Level			*	*	*